MILITARY SPECIFICATION

SWAGING MACHINE, AIRCRAFT CABLE

PORTABLE, POWER OPERATED, MODEL NSM-1

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers one type of portable, power operated, swaging machine to be used for the swaging of terminals on aircraft control cables.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
TT-E-529	Enamel; Alkyd, Semi-Gloss
UU-P-271	Paper, Wrapping, Waterproofed Kraft
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
TT-C-490	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

Military

MIL-P-116	Preservation, Method of
MIL-C-1511	Wire Rope, Steel (Carbon) Flexible, Preformed

FSC 3450

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SPECIFICATIONS (Cont'd)

Military

MIL-C-4109	Coupling Assembly, Air-Hose, Quick Operating
MIL-C-5424	Cable; Steel (Corrosion-Resisting), Flexible, Preformed (For Aeronautical Use)
MIL-C-5693	Wire Strand, Steel (Corrosion-Resistant), Preformed (Aircraft Applications)
MIL-T-6117	Terminal, Cable Assemblies, Swaged Type
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-P-18058	Preparation for Delivery of Machine Tools for Domestic and Overseas Shipment

STANDARDS

Federal

FED-STD-595	Colors (For), Ready Mixed Paints

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

Air Force - Navy Aeronautical

MS-20658	Terminal, Wire Rope, Swaging, Fork End (for Bearings)
MS-20663	Ball End, Wire Rope, Swaging, Double Shank
MS-20664	Ball End, Wire Rope, Swaging, Single Shank
MS-20667	Terminal, Wire Rope, Swaging, Fork End

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STANDARDS (Cont'd)

Air Force - Navy Aeronautical

MS-20668	Terminal, Wire Rope, Swaging, Eye End
MS-21259	Terminal, Wire Rope, Stud
MS-21260	Terminal, Wire Rope, Stud

(When requesting specifications, standards, drawings and publications refer to both title and number. Copies of applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue Philadelphia, Pennsylvania 19120 Code 105)

2.2 Other Publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Material - Materials used in the manufacture of the swaging machines shall be of high grade quality, selected to provide a long maintenance free service life and be entirely suitable for the intended purpose.

3.1.2 Standard Parts - Standard parts shall be used wherever they are suitable for the purpose. Commercial utility parts such as screws, bolts, nuts, etc., may be employed if they possess suitable properties and are replaceable by Standard (MS, AN or JAN) parts without alterations.

3.2 Design and Construction - The swaging machine shall be designed and constructed so as to provide a portable, reliable and convenient means of swaging aircraft terminals and for cutting aircraft control cable. The machine shall be capable of swaging terminals in accordance with paragraph 3.1 of Specification MIL-T-6117. The unit will be capable of being operated by a remote foot control.

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3.2.1 Cutting Capacity - The machine shall be capable of cutting steel cable conforming to Specifications, MIL-C-1511, MIL-C-5424 and MIL-C-5693 in all sizes up to and including 1/4 inch. The cut end shall not be ragged or unraveled and shall be ready to be inserted into a terminal for swaging.

3.2.2 Swaging Capacity - The machine shall be capable of swaging ball shank terminals conforming to Standards MS20663 and MS20664 in all sizes of cable up to and including 3/16 inch and sleeve type terminals conforming to Standard MS20658, MS21259, MS20667, MS20668 and MS21260 in all sizes of cable up to and including 1/4 inch. The cable shall conform to Specification MIL-C-1511, MIL-C-5424 and MIL-C-5693.

3.2.3 Swaged Terminals - Swaged terminals produced by the swaging machine shall conform to the dimensions specified in the applicable standard, and the distortion and tensile strength requirements of paragraphs 3.4 and 3.5 of Specification MIL-T-6117. The machine shall also be capable of trimming the cable on ball type terminals as specified by Standard MS20664. The swaged terminals shall have a smooth surface free from die marks and rapid dimensional changes in cross section; shall not become brittle as a result of cold working and shall not contain splits, cracks or other injurious defects.

3.2.4 Swaging Time - The time to swage any terminal shall not exceed 3 minutes at 90 psi air pressure.

3.2.5 Portability - The machine shall be so designed that it may be operated both at a work bench or inside of an aircraft. Means shall be provided whereby the machine, dies and accessories may be readily carried from one job to another.

3.2.5.1 Weight - The weight of the swaging machine shall not exceed 30 pounds. The weight of the machine together with all swaging dies, accessories and carrying case shall not exceed 90 pounds.

3.2.6 Power - The machine shall be pneumatically, operated and shall be capable of operation with air pressures ranging from 70 to 120 pounds per square inch.

3.2.6.1 Air Inlet - The air inlet connection shall have female 1/4 inch pipe threads. A quick operating, nipple conforming to Specification MIL-C-4109 shall be provided for attaching the swager to the air supply hose.

3.2.6.2 Strainer - The swager shall be fitted with a 50 micron or less air strainer or a 200 mesh or finer screen which shall be effective in retaining solid particles in the compressed air supply.

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3.2.6.3 Foot Control - The swaging machine shall be equipped with a foot control incorporating a single action on-off type valve. The foot control shall be enclosed in a heavy duty steel frame and shall be so designed as to prevent inadvertent actuation of the valve control pedal.

3.3 Dies and Accessories - All dies and accessories necessary for the swaging and cutting operations of the capacity specified in paragraph 3.2, 3.2.1 and 3.2.2 shall be provided.

3.3.1 Gage - Suitable gages shall be provided with the swaging machine to determine the acceptability of the swaged terminals. The gage shall be of the go-no-go type, the openings shall be in accordance with the dimensional tolerances specified inthe applicable Standard, MS20658, MS20663, MS20664, MS20667, MS20668 and MS21260. The gages shall be manufactured from tool steel, hardened and ground to resist wear.

3.3.2 Carrying Case - The carrying case shall be manufactured from at least 24 gage steel. The case shall have specific compartments for the swaging machine, dies, gages, accessories and other loose parts. At least two (2) handles shall be provided for carrying the case.

3.4 Maintenance and Service - The swaging machine shall be designed and constructed so that no parts will work loose in service. It shall withstand the strains, jars, vibrations and other conditions incident to shipping storage and service and shall require only a minimum of maintenance. Such maintenance when required shall be accomplished without the need of special tools or contractor provided services. All parts subject to replacement or adjustment shall be easily and readily accessible.

3.5 Interchangeability - All replaceable parts shall be manufactured to definite commercial standards, tolerances and clearances in order that any such parts may be replaced without requiring modification. Where practicable such parts shall be permanently and legibly marked with the applicable drawing number.

3.6 Threads - All threaded parts shall be in accordance with the applicable requirements of Handbook H28.

3.7 Fastening Devices - All screws, pins, bolts and similar parts shall be installed with adequate means for preventing loss of proper tightness or adjustment. All such parts when subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently deformed.

3.8 Lubrication - The swaging machine shall be equipped with an

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automatic feed reservoir type oiler capable of supplying lubricant to all parts requiring lubrication. The unit shall be capable of being operated efficiently and satisfactorily with a standard military lubricant.

3.9 Safety - All rotating and reciprocating parts shall be fully enclosed or properly guarded.

3.10 Identification of Product - The swaging machines shall be marked for identification in accordance with paragraph 4.3.1 of Standard MIL-STD-130.

3.11 Painting - Metal surfaces requiring painting, shall be cleaned and prepared in accordance with Specification TT-C-490, , and shall be painted with at least one coat of primer conforming to Specification MIL-P-8585 followed by two coats of enamel conforming to Specification TT-E-529. The color shall be yellow conforming to Color Number 23538 of Standard FED-STD-595.

3.11.1 Plating - All unpainted steel parts except dies shall be plated in accordance with Specification QQ-P-416.

3.12 Workmanship - The swaging machine, dies, gages and accessories shall have a smooth finish, free from burrs, fins, flows, and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conforming to prescribed requirements.

4.2 Preproduction Inspection and Test - One swaging machine shall be submitted to the Government for preproduction inspection and tests. The unit submitted shall be identical in design, quality of workmanship and material to those the manufacturer intends to supply. The inspection and tests are intended to determine complete compliance with the requirements of this specification. Fabrication of any additional swagers or components thereof prior to approval of the preproduction model, shall be at the contractor's risk and such additional swagers or components will be subject to rejection in the event they fail to completely conform to this specification.

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4.2.1 Preproduction Inspection - The preproduction model shall be thoroughly examined to determine compliance with the requirements of this specification.

4.2.2 Preproduction Tests - The preproduction model shall be subjected to the following tests.

4.2.2.1 Test Cable - At least one test cable shall be made up from each size and type cable and each size and type terminal listed in paragraph 3.2.2 hereto.

4.2.2.2 Dimension Tests - The swaged terminals shall then be thoroughly examined to determine that they comply with the dimensional requirements specified in the applicable Standard, MS 20658, MS 20663, MS 20664, MS 20667, MS 20668 and MS 21260.

4.2.2.3 Distortion Test - Applicable cable assemblies shall be subjected to and meet the requirements of the distortion test specified by Specification MIL-T-6117.

4.2.2.4 Tensile Strength - The test cable assemblies shall be subjected to and meet the requirements of the tensile strength test specified by Specification MIL-T-6117.

4.2.2.5 Visual Test - The terminals on the test cable assemblies shall be thoroughly examined to determine that the surface is free from damaging die marks, that no rapid dimensional changes in cross section are evident, that the terminal has not become brittle from cold working and that no splits, cracks or other injurious defects are present.

4.2.3 Life Test - The swaging machine shall be operated under actual service conditions for a period of 100 hours. At the end of this test the unit shall be inspected for vibration, loose parts and operation.

4.2.4 Air Pressure - The above tests shall be performed on pneumatic machines at an air gage pressure of 90 pounds per square inch measured at the end of the hose attached to the pressure gage.

4.2.5 Reinspection and Retest - Units not capable of passing the foregoing tests shall be rejected. Rejected units shall not be submitted for reinspection or retest without the contractor first furnishing full particulars concerning previous rejections and measures taken to correct the defects.

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4.3 Production Inspection - Materials, components, sub-assemblies and parts entering into the manufacture of the swaging machines shall be inspected during production and assembly to determine that they duplicate the production model and conform to this specification.

4.4 Acceptance Test and Inspection - At least one swaging machine shall be selected at random from every lot of 10 offered for acceptance and subjected to the following acceptance inspection and testing. When one swaging machine selected from the production run fails to meet the specification, no swaging machines on hand or later produced shall be accepted until the extent and cause of failure are determined and corrected.

4.4.1 Acceptance Inspection - The approved preproduction model shall be used as a standard for the inspection of the production units. The production units selected as per paragraph 4.4. above shall be thoroughly examined to determine that it is a true duplicate of the preproduction model and conforms to the requirements of this specification.

4.4.2 Acceptance Test - The acceptance unit selected as per paragraph 4.4 above shall be subjected to the tests specified by paragraphs 4.2.2.1, 4.2.2.2, 4.2.2.3, 4.2.2.4 and 4.2.2.5 hereto, except that only test cables manufactured from cable conforming to Specification MIL-C-1511 need be used.

4.4.3 Examination of Preparation for Delivery - Inspection to determine compliance with preparation for delivery requirements shall be accomplished in accordance with MIL-P-18058.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Swaging machines and accessories shall be preserved and packaged in accordance with Level C of MIL-P-18058 (see 6.2).

5.1.2 Overseas Shipment - Swaging machines and accessories shall be preserved and packaged in accordance with Method IIA of Specification MIL-P-116.

5.2 Packing -

5.2.1 Domestic Shipment - Swaging machines preserved and packaged as specified by 5.1 shall be packed in a manner that will insure acceptance and safe transportation at the lowest rate to the point of delivery.

5.2.2 Overseas Shipment - Swaging machines preserved and packaged

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as specified by 5.1.2 shall be wrapped in waterproof paper conforming to Specification UU-P-271 and overpacked in overseas-type shipping containers conforming to Specifications PPP-B-591 or PPP-B-621.

5.3 Marking of Shipments - Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended Use - The swaging machine covered by this specification is intended for use in attaching terminals to aircraft control cables.

6.2 Ordering Data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Preservation, packaging and packing, if other than as specified in 5.1.

(c) Special marking (see 5.3).

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-S-81353(WP) SWAGING MACHINE, AIRCRAFT CALBE PORTABLE, POWER OPERATED, MODEL NSM-1

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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